@Question3

```
START
Function input(x, y, arr)
    for i=0 to x-1
        for j=0 to y-1
            arr[i][j] ← input
        End for
    End for
End function
Function Calc(x, y, arr)
    Declare total, high, low, critical, warning, avg, streak, longest.
    for i=0 to x-1
        set total ← 0
        set avg ← 0
        set critical ← 0
        set warning ← 0
        set high ← arr[i][0]
        set low ← arr[i][0]
        set streak ← 0
        set longest ← 0
        for j=0 to y-1
            total ← total + arr[i][j]
            if arr[i][j] > high
                high = arr[i][j]
            End if
```

Scanned with CamScanner

```
if arr[i][j] < low
    low = arr[i][j]
Endif.
if arr[i][j] >= 40
    critical ← critical+1
Endif
if arr[i][j] >= 35 AND arr[i][j] <= 39
    warning ← warning+1
Endif.
If arr[i][j] >= 35
    streak ← streak+1
    if streak > longest
        longest = streak.
    end if.
    else
    streak ← 0
    end if.
endfor
avg ← total / y
print Sensor number, total, avg, high, low, critical,
warning, ~~longest~~ longest
endfor.
end function.
```

Scanned with CamScanner

```
function highestAverage(x,y,arr)
    Set sens ← 0
    Set havg ← 0
    for i ← 0 to x-1
        set total ← 0
        for j ← 0 to y-1
            total ← total + arr[i][j]
        end for
        set avg ← total / y
        if avg > havg
            havg = avg
            sens = i
        endif
    end for
    return sens.
End function.
Main
    Declare sens, temp, total, avg.
    Set total ← 0
    Set avg ← 0
    sens ← input
    temp ← input
    declare arr[sens][temp]
    call input(sens, temp, arr)
```

Scanned with CamScanner

Call calc(sens, temp, arr)

Set sensor ← highest Average (sens, temp, arr)

for j=0 to temp-1

total ← total + arr[sensor][j]

End for.

avg ← total/temp

print sensor with highest avg → sensor+1

print highest average temp → avg

End.

Input :-

- Number of sensors
- Number of temperature readings.

Output :-

- total temp for each sensor.
- Avg temp for each sensor
- Highest and lowest for each sensor.
- Number of critical readings
- number of warning readings
- streaks
- Sensor with highest average temperature
- highest average temperature